Assignment Handout

Competitive Data-Center Connection Planning Game

Course name

October 6, 2026

Assignment in one sentence. You operate a data-center, choose where and how much load to connect in a power grid, optionally build private generation and storage, and maximize your profit over a six-month hourly operating horizon.

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1 Assignment overview

Each student group represents a data-center operator such as a cloud-computing company. Your group submits a connection-and-resource plan. Submissions of all groups will be collected and evaluated together using a DC optimal power flow (`DCOPF`) dispatch model. **Groups are ranked by score.**

1.1 What you must do

Your team must design and submit one complete half-year connection plan in the released CSV format. The goal is to serve valuable computational load while controlling energy, interconnection, generation, and storage costs.

1.2 Deliverables

1. one `UTF-8 CSV` file for your team;
2. validator output showing that the file passes the selected `horizon`;
3. a short strategy note explaining your main bus, load, generator, and storage choices and the trade-offs you considered;
4. the exact final CSV and any code used to generate it.

The instructor will announce the file name, submission location, deadline, round assignment, and whether the strategy note is required for grading. Do not submit a screenshot in place of the CSV.

1.3 Assignment rules at a glance

- Submit exactly one complete `half-year horizon`: 4,368 hours for January–June or 4,416 hours for July–December.
- Use `bus IDs` and `generator IDs` from the supplied Zenodo artifact.
- Load is nonnegative and is specified directly in `MW` for every hour.
- `Generators` and `storage` are private to the submitting team.
- The current release has no student-controlled `curtailment` decision.
- The instructor evaluates valid teams jointly and ranks teams by `score`.

Use of AI. You are explicitly encouraged to use available AI tools while doing the assignment, including ChatGPT, Codex, Claude Code, GitHub Copilot, and similar tools. You remain responsible for checking generated code, understanding your submission, respecting the CSV contract, and explaining your engineering decisions. AI assistance is part of the intended computational workflow, not a substitute for validation.

1.4 Learning objectives

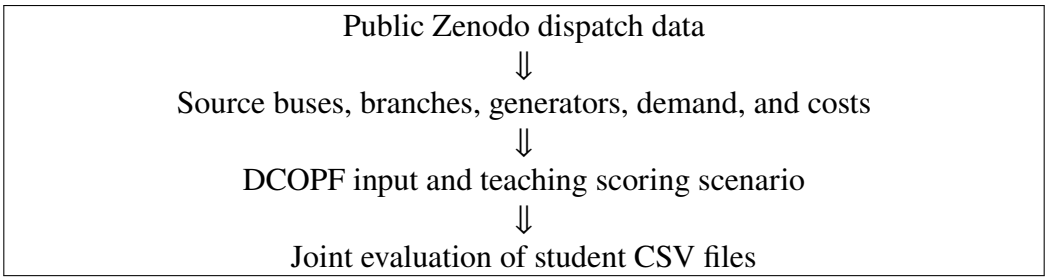
- translate a planning idea into a machine-readable decision file;
- reason about spatial congestion, generator dispatch, and storage;
- understand why individually feasible plans can be jointly infeasible;
- use reproducible computational experiments and inspect diagnostics;
- communicate trade-offs among IT value, energy cost, and investment.

2 Data and model: what is real and what is instructional

The dispatch input is the Texas portion of the U.S. Test System dataset released on `Zenodo` by Xu et al. [9]. The prepared Texas subset used contains approximately **2,000 buses** and hourly demand and network data.

The prepared demand artifact is timestamped 2016. The course uses its January–June and July–December halves as two teaching-round calendars; this is a historical replay for the exercise, not a 2028 forecast.

The conceptual pipeline is:



The following distinction is important:

- **Public data:** `bus`, `branch`, `generator`, `demand`, and marginal-cost records from the Zenodo artifact.
- **Instructor preparation:** `formatting`, `validation`, and the scoring vectors supplied with the round.
- **Instructional assumptions:** the student-owned investments, connection requests, `DCOPF` treatment, `IT value`, capital-cost coefficients, storage-cost coefficients, and the `fair continuous projection` policy.

The exercise should not be interpreted as an ERCOT planning recommendation or as a forecast of actual interconnection feasibility.

3 Operating horizon and decisions

Each submission governs **one half-year with a one-hour time step:**

Round	Hours	Zero-based period indices
January–June	4,368	$0, \ldots, 4367$
July–December	4,416	$0, \ldots, 4415$

The full source year contains 8,784 hourly records. A submission therefore has access to roughly 2,000 possible buses and roughly 4,500 hourly decision slots, but it does *not* need to contain 2,000×4,500 CSV rows. The long-form CSV lists only the buses used by your team. For every bus that appears in the file, however, the release evaluator requires one row for every hour in the selected half-year; use a zero load in a row when that bus is idle. Buses that never participate may be omitted entirely.

For each selected `bus` s and hour t, you may request load $d_{s,t} \geq 0$. You may also submit one or more private `generators` and one or more private `storage devices` attached to your selected connection sites. A team's resources serve only that team. They cannot export energy to another team or become a shared pool.

4 Mathematical model

4.1 Connection load

Let $d_{i,s,t} \geq 0$ be the load requested by team i at bus s in hour t. The corresponding connection capacity is

$$C_{i,s} = \max_t d_{i,s,t}, \qquad p_{i,s,t} = \begin{cases} d_{i,s,t}/C_{i,s}, & C_{i,s} > 0, \\ 0, & C_{i,s} = 0. \end{cases}$$

The CSV stores the hourly load directly; the evaluator reconstructs the capacity and normalized profile internally.

4.2 Private generation

`Generator IDs` refer to records in the Zenodo generator catalog. The evaluator uses the catalog's `marginal_cost_c1_per_mwh` value for the generator's operating cost. The generator is physically attached to your chosen bus for the exercise, but its cost reference comes from the public dataset. For a generator with capacity $G_{i,g}$, dispatch satisfies

$$0 \leq g_{i,g,t} \leq G_{i,g}.$$

The instructor may impose additional catalog or site limits in the round configuration.

For the current classroom release, the dispatch evaluator uses the catalog `nameplate limit` for each selected generator. The prepared Zenodo hourly renewable profile files are retained as source metadata but are not imposed as hard availability caps in this no-curtailment version of the exercise.

4.3 Private storage

For storage device b, submit power capacity $P_{i,b}$ in MW and energy capacity $B_{i,b}$ in MWh. Charging $c_{i,b,t}$ and discharging $q_{i,b,t}$ are continuous. We assume ideal charging/discharging efficiency:

$$e_{i,b,t} = e_{i,b,t-1} + c_{i,b,t} - q_{i,b,t},$$

with

$$0 \le e_{i,b,t} \le B_{i,b}, \qquad 0 \le c_{i,b,t}, q_{i,b,t} \le P_{i,b},$$
$$q_{i,b,t} \le e_{i,b,t-1}, \qquad e_{i,b,0} = 0.5B_{i,b}, \qquad e_{i,b,H} \ge 0.5B_{i,b}.$$

Storage is private to the submitting team. Charging appears as additional team demand and discharging reduces that team's grid purchase; it does not serve any other team's load.

Storage cost calibration. The released half-year scoring files use the following instructional coefficients:

$$C^{\text{storage,cap}}_{i,b} = (\$200{,}000/\,\text{MW})P_{i,b} + (\$300{,}000/\,\text{MWh})B_{i,b},$$
$$C^{\text{storage,op}}_{i,b} = (\$8/\,\text{MWh}) \times \text{discharged energy}.$$

For a four-hour device, the two capital terms correspond to approximately \$1,400/kW of installed power capacity. These are rounded teaching coefficients informed by U.S. battery-storage construction and O&M benchmarks, not a project quotation or an ERCOT tariff [8, 5, 4]. The same coefficients are used for both half-year rounds unless the instructor announces a revised round file. The reference literature also reports fixed O&M near \$40/kW-year; the current evaluator does not have a separate annual fixed-O&M field, so that value is documented as a reference and is not charged in addition to the \$8/MWh operating proxy.

4.4 DCOPF

For each hour, the instructor solves a linear `DCOPF` on the prepared Texas network. Network balance, generator limits, branch thermal limits, and fixed base demand are enforced. Load is inelastic in the ordinary feasible case: there is no integer unit commitment and no ordinary load-shedding variable.

5 CSV submission format

Submit one `UTF-8 CSV` file per team. The canonical header is:

```
team_id,submission_id,period,bus_id,load_mw,generator_id,generator_capacity_mw,
    storage_id,storage_power_mw,storage_energy_mwh
```

Each row contains a one-based hour `period` and a Zenodo `bus_id`. A `generator` or `storage` specification may be repeated on rows for the same team; its capacity and attachment must remain consistent. The load for a given bus–period must also remain consistent across repeated resource rows.

For a released half-year submission, each selected bus must cover every period at least once; additional rows may declare distinct generator or storage resources, provided the load and resource specifications remain consistent. The short table below illustrates the column format only; it is not a complete half-year submission. A file that contains only the illustrated rows will be rejected when checked with `-horizon 4368` or `-horizon 4416`.

Example:

```
team_id,submission_id,period,bus_id,load_mw,generator_id,generator_capacity_mw,
    storage_id,storage_power_mw,storage_energy_mwh
team-blue,half_year_1,1,3002041,40,12875,20,BATT_1,10,20
team-blue,half_year_1,2,3002041,35,12875,20,BATT_1,10,20
team-blue,half_year_1,1,3002042,25,,,,,
```

Use empty fields for resources that are not present on a row. Values use MW, MWh, and decimal notation. Periods are one-based in the CSV even though the internal evaluator uses zero-based indices.

The `validator` also checks complete horizon coverage. This prevents a short pilot file from silently becoming a mostly-zero half-year plan. The released checker accepts the two course horizons only: use 4368 for January–June or 4416 for July–December.

Before submitting, run the provided `CSV validator`. From the project root, run (replace the artifact path with the released Zenodo artifact path):

```
PYTHONPATH=src python scripts/check_submission.py \\
  --file team-blue.csv \\
  --artifact-dir data/processed/texas_dispatch_source_v1 \\
  --horizon 4368
```

Use `-horizon 4416` for the July–December submission. The `checker` is also available after installation as `ercot-game-check-csv`. It checks UTF-8 encoding, the exact header, numeric fields, bus and generator IDs, duplicate keys, consistent resource specifications, horizon bounds, and nonnegative values. The validator is a convenience check; the instructor evaluator remains authoritative.

5.1 Complete small example

The following illustrates a valid two-hour pattern, but it is not a valid course submission because it is shorter than either released horizon:

```
team_id,submission_id,period,bus_id,load_mw,generator_id,generator_capacity_mw,
    storage_id,storage_power_mw,storage_energy_mwh
team-demo,demo-1,1,3002041,4,12875,2,BATT_1,1,2
team-demo,demo-1,2,3002041,3,12875,2,BATT_1,1,2
team-demo,demo-1,1,3002042,2,,,,,
team-demo,demo-1,2,3002042,0,,,,,
```

The resource specifications are repeated consistently for the first bus. The second bus has no generator or storage, so those fields are empty. In an actual submission, this coverage pattern continues through period 4,368 or 4,416.

5.2 Checker troubleshooting

- **Unknown bus or generator:** copy the ID from the supplied Zenodo catalog; do not invent a new ID.
- **Incomplete horizon:** add rows for every missing period on each selected bus, using zero load when the bus is idle.
- **Inconsistent resource specification:** repeat the same capacity and attachment values on every row for that resource.
- **Malformed CSV:** save as UTF-8 CSV and inspect commas, quotation marks, and the exact header.

6 How joint evaluation works

The instructor first rejects `malformed CSV` files in isolation; a malformed file does not prevent other teams from being evaluated. Review the exact file contract in Section 5 before diagnosing a rejection.

Valid plans are then evaluated jointly. If the joint plan is `feasible`, each team is scored using its cleared hourly load, private resource dispatch, and the announced scoring vectors.

If the joint plan is `infeasible`, the evaluator applies the `fair continuous projection`, discussed in Section 7. No first-come-first-serve ordering is used. The projected load is then used for the score. The capital costs associated with submitted sites, generators, and storage remain part of the score. See Section 8 for the score calculation.

7 Fair continuous projection

Let $d_{i,s,t}$ be the requested load of team i at bus s at time t. When the full joint request is infeasible, the evaluator finds a common continuous service fraction $\alpha \in [0, 1]$ and evaluates the scaled requests

$$\widehat{d}_{i,s,t} = \alpha d_{i,s,t}.$$

The largest feasible α is selected to *maximize the minimum fraction of each team's requested plan that is served.* This policy has three intended properties:

1. every valid team remains in the competition;
2. teams are treated symmetrically at the service-fraction level;
3. the projected dispatch remains a physically feasible DCOPF solution.

The policy is related to `max-min` and proportional fairness ideas used in resource allocation and demand-response research [3, 2, 1]. It is a teaching rule, not an ERCOT operating procedure.

8 Scoring

For team i, the `score` is

$$S_i = V_i^{\mathrm{IT}} - C_i^{\mathrm{energy}} - C_i^{\mathrm{gen,op}} - C_i^{\mathrm{gen,cap}} - C_i^{\mathrm{storage,op}} - C_i^{\mathrm{storage,cap}} - C_i^{\mathrm{site}}.$$

The components are:

- V_i^{IT}: cleared load multiplied by the hourly `IT value` vector announced for the round;
- C_i^{energy}: grid purchase multiplied by the hourly `energy price` price vector;
- $C_i^{\mathrm{gen,op}}$: private-generator dispatch multiplied by the dataset-derived marginal cost;
- $C_i^{\mathrm{gen,cap}}$: submitted generator MW multiplied by the announced generator capital coefficient;
- $C_i^{\mathrm{storage,op}}$: announced `storage operating coefficient` multiplied by storage throughput;
- $C_i^{\mathrm{storage,cap}}$: announced storage MW/MWh capital coefficients multiplied by submitted capacities;
- C_i^{site}: number of submitted `connection sites` multiplied by the announced site coefficient.

The current provisional calibration anchors are listed below. The instructor must distribute the final hourly vectors and storage coefficients with the round configuration; do not infer them from this manual if the round file differs.

Parameter	Provisional value	Basis
Generator capital	$85,000/MW	Annualized combustion-turbine anchor from EIA cost data [7].
Site cost	$50,000/site	Instructional interconnection proxy informed by Berkeley Lab interconnection-cost work [6].
Generator operating cost	Dataset value	Zenodo `marginal_cost_c1_per_mwh` record.
Energy price	Round vector	Instructor-selected ERCOT DAM reference series; source documentation is supplied with the round.
IT value	Round vector	Instructional value-of-compute assumption; sensitivity-tested by the instructor.
Storage power capital	$200,000/MW	Rounded power-capacity component calibrated from NREL/EIA battery benchmarks [8, 4].
Storage energy capital	$300,000/MWh	Rounded energy-capacity component; together with power capital gives about $1,400/kW for a four-hour system.
Storage operation	$8/MWh discharged	Variable O&M proxy; the reference fixed O&M value is $40/kW-year [8].
Invalid file score	0	Deterministic grading policy, not a market quantity.

8.1 How to reason about your score

The evaluator first determines how much of your requested load is served. It then adds IT value for served load and subtracts energy, site, generator, and storage costs. A larger plan is therefore not automatically better: an additional site may increase IT value while also increasing energy, congestion, and capital costs. A generator or storage device can reduce grid purchases but still lower the final score if its costs exceed the avoided energy cost.

Use the hourly `price` and `IT-value` vectors supplied with the round when estimating a score; do not replace them with a guessed constant price.

9 Practical workflow

1. Read the round configuration and identify the horizon.
2. Inspect the Zenodo bus and generator tables supplied by the instructor.

3. Choose one or more buses and build hourly load profiles.
4. Decide whether private generation or storage improves your score.
5. Generate the `CSV` and run the `validator`.
6. Keep the exact submitted file and validator output for reproducibility.
7. Submit before the announced deadline.

9.1 What to inspect after evaluation

Read the result in this order:

1. feasibility or projected service status;
2. total score and rank;
3. cleared load, grid purchases, and resource costs;
4. binding or limiting network diagnostics;
5. storage state and generator dispatch;
6. raw audit output only when debugging.

9.2 Final submission checklist

Before uploading, confirm:

- the file is `UTF-8 CSV` with the exact header;
- every selected bus has one row for every required period;
- all bus and generator IDs come from the supplied catalogs;
- generator and storage specifications are consistent across rows;
- all loads and capacities are finite and nonnegative;
- the `checker` passes with the correct half-year horizon;
- you retained the exact submitted file and checker output;
- your strategy note explains the main economic and physical trade-offs.

10 Troubleshooting and frequently asked questions

Can I invent a generator ID or choose its operating cost?
No. `Generator IDs` and operating costs come from the supplied Zenodo catalog. You choose capacity; the evaluator dispatches the generator.

Can my generator or storage serve another team?
No. These resources are private and can reduce only your team's grid purchase.

What happens if the joint plan is infeasible?
The evaluator first applies the fair continuous projection in Section 7, using one common service fraction for valid teams.

Why is my score negative?
Served-load value may be smaller than energy, site, generator, or storage costs. Inspect the detailed score components before changing the plan.

Which result is official?
The instructor-generated result and audit files determine the official score. Local calculations are for debugging and strategy development.

11 References

References

[1] Dalia Gebbran et al. Fairness-aware coordination of distributed energy resources. *IEEE Transactions on Smart Grid*, 2021. Reference for continuous proportional and egalitarian resource allocation.

[2] Doron Jacubowicz, Guy Raz, and Yair Beck. Fairness algorithm for emergency demand response. *International Journal of Electrical Power and Energy Systems*, 134:107871, 2022.

[3] Frank P. Kelly. Charging and rate control for elastic traffic. *European Transactions on Telecommunications*, 8(1):33–37, 1998.

[4] National Renewable Energy Laboratory. Annual technology baseline 2024: Utility-scale battery storage, 2024. Reference for separating battery power and energy capital costs.

[5] Vignesh Ramasamy et al. Annual technology baseline: Utility-scale battery storage. Technical report, National Renewable Energy Laboratory, 2021. Reference for utility-scale four-hour battery capital-cost components.

[6] Joseph Rand et al. Generator interconnection costs to the transmission system. Technical report, Lawrence Berkeley National Laboratory, 2022.

[7] U.S. Energy Information Administration. Capital cost and performance characteristics for utility-scale electric power generating technologies, 2024. Technology-cost reference for the provisional generator capital anchor.

[8] U.S. Energy Information Administration. Capital cost estimates for utility scale electricity generating plants: Battery energy storage system assumptions, 2024. Reference for battery construction, fixed O&M, and variable O&M benchmarks.

[9] Q Xu et al. U.s. test system, 2021. Public research dataset used for the Texas dispatch artifact.